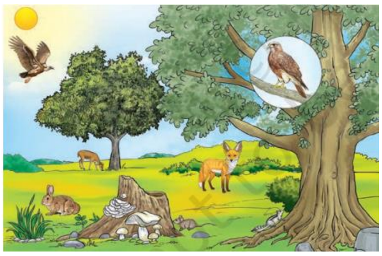
Chapter – 12: How Nature Works in Harmony

- Several parts – India – particular states – Odisha, Jharkhand, West Bengal, Assam, Chhattisgarh – elephants – enter farms, villages
- Vegetation – scarce – dry waterholes – natural habitat – elephants – wander – nearby farms – search food – bananas, sugarcane
- Lead – crops damage – sometimes – harm people, domestic animals
- Changes – rainfall, temp. – affect vegetation – cutting trees – construction – roads, buildings – made worse
- Lead – drying, shrinking forest – natural home – animals
- Loss of habitat – animals – move – human habitats
- Elephants – adapted – forest life – BUT – sudden changes – hard – survive
- Wildlife ecologists – identify, mark corridors – many parts – country – safe movement – animals
- Corridor – connect – forest habitats – enable wildlife – elephants – travel – large forest areas – no conflict – human settlements
- Chain of events – show – nature's close connection
- Understand – interconnection – study – components – env.

How Do We Experience and Interpret Our Surroundings?

- Learnt – curiosity – grade 6 – chapter – diversity – diff. habitats – diff. kind – plants, animals
- Habitat – place – organism live – plants, animals – interact – each other – adapt – survive – surrounding cond.
- **Explore** – 2 habitats – **identify** both – living, non-living components

- Activity –
    - Identify – 2 habitats – surroundings
    - Any of the following – pond, forest, farm, tree – banyan, mango, *pilkhan* (white fig)
    - List – living beings – non-living things – **observe** – habitats
    - **Record** – obs. – table



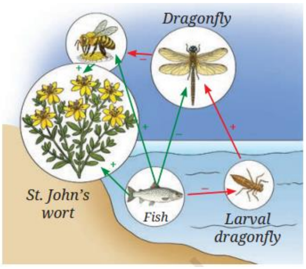
- Common characteristics – obs. – 2 habitats – above act.
- Similarities – both habitats – living, non-living things – BUT – types – living, non-living things – vary
- Living beings – above table – termed – **biotic** – non-living – **abiotic**
- Some org. – land – others – water – every org. – specific cond. – survive
- This act. – visible – diff. habitats – diff. living cond.
- Above act. – listed fish – biotic component – pond
- Pond – food, oxygen, shelter, space – grow – survive
- Fish – biotic needs – food – plants, animals – abiotic needs – oxygen – water
- Other living beings – inhabit pond – frogs, turtles, snakes, dragonflies, mosquitos, snails, ducks – plants – algae, diatoms, duckweeds, lotus
- All – interact – other living, non-living things – present – places – grow, thrive
- Each habitat – own components – phy. cond. – air, sunlight, water, temp., soil
- Diff. org. – live – same habitat – use resources – diff. ways
- Forest – warm – day – cool – night
- Snake – comes out – night – rodent – active – day – both – same habitat – diff. cond.

- This way – living org. – coexist – harmony – same habitat

Who All Live Together in Nature?

- Obs. fish – pond – above act. – many fish – same kind
- Grp. – fish – same kind – live together – pond habitat – **population**
- This way – obs., record populations – diff. kinds – org. – single habitat
- Activity –
  - Understand population – particular plant – count them – given place, time
  - Divide std. – 4, 5 grps.
  - Each grp. – identify – 2 org. – plants, animals
  - Mark area – 1 m x 1m – school garden
  - Identify – 4 org. – count no. – this area
  - Record obs. – compile data – all grps.
- Given ex. – population – plant 1 – 20 – plant 2 – 05 – same 1 x 1 m^2 area
- This act. – explain – population – grp. – same type – org. – habitat – given time
- All org. – same – same req. – food, water, space – lead – competition, scarcity – resources
- Above act. – obs. – diff. grp. – org. – live together – habitat
- **Community** – diff. population – share same habitat
- Biotic components – habitat – plants, animals, microorganisms – together – comm. – interact, depend – one another – survival
- **Interesting fact** –
  - Bright colour flowers – bloom around – visible
  - Flower – stalk, green leafy structures – **sepals** – coloured **petals** – 2 reproductive parts – **carpels** (female), **stamen** (male)
  - Stamens – burst – release yellow dust – **pollen grains**
  - Wind, water, insects, bats, birds – carry pollen – stamen to carpels – same and/or diff. flowers
  - This process – **pollination** – essential – formation – fruits, seeds

Does Every Organism in a Community Matter?



- Activity –
  - Researchers – **conduct** study – fish – pond – affect – seed production – nearby plants
  - Obs. 2 ponds –
    - A – filled – fish – large no. – flowering plants
    - B – empty – fewer flowering plants
  - **Compare** no. – dragonflies, bees, butterflies – obs. – pond A – dragonflies – lesser no.
  - Fish eat larvae – ponds – fish – fewer dragonflies – eat less flies, bees, butterflies – result – more bees, flies, butterflies – pollinate flowers – nearby areas – help plants – produce seeds – flowers – ponds – fish – more seeds
  - This study – show – biotic, abiotic – interact, affect – each other – overfishing – change balance

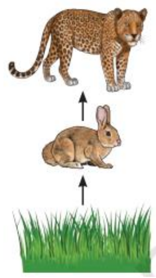
What Are the Different Types of Interactions Among Organisms and their Surroundings?

- Curiosity – grade 7 – learn – plants, animals – need air, water, soil, sunlight – grow
- Biotic comm. – depend – abiotic components – survival

- Plants, animals – also depend – each other – nutrition, respiration, reproduction
- Both types – interactions – inter-biotic, biotic-abiotic – imp. – survival – any habitat
- Activity –
  - Based – criteria – **identify**, **describe** – interactions – biotic, abiotic
  - Criterion 1
    - Interactions – biotic, abiotic components – influence – life processes – nutrition, respiration, reproduction – biotic components
  - Criterion 2
    - Interactions – 2 abiotic components – influence – phy. characteristics – habitat
  - Criterion 3
    - Interactions – biotic components – influence availability – resources – life processes – nutrition, respiration, reproduction
  - **Relate** – learning – obs. – record – table

Criterion 1 : interactions – biotic, abiotic components	**Criterion 2 : interactions – 2 abiotic components**	**Criterion 3 : interactions – biotic components**
Earthworms – live – moist soil	Day temp. – high – bright sunlight	Frog – eat insect
Microbes – present – pond	Water – evaporate fast – sunlight	Water snake – eat fish
Fish – lay egg – water	Air current – blow slow – water surface – gentle waves	Frogs, fish – compete – insect larvae
	Soil – near pond – moist	Fish – lay eggs – water – vegetation – protect – other fish, frogs

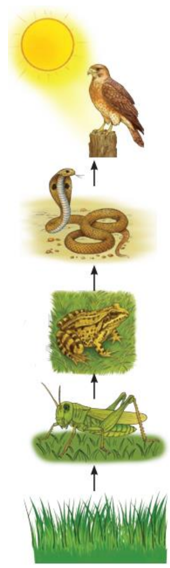
- This act. – understand – diff. interactions – habitat
- This – infer – biotic, abiotic components – habitat – interact – each other – form **ecosystem**
- Organisms – ecosystem – interact – abiotic components – food, shelter, protections
- Diff. comm. – living org. – interact – abiotic components – ecosys.
- Nature – 2 main ecosys. –
  - **Aquatic ecosystems** – ponds, rivers, lakes
  - **Terrestrial ecosystems** – forests, farms, large trees – banyan, mango, *pilkhan*
- Ecosystems – large, small – sometimes – overlap
- Fig. – overlap – diff. terrestrial, aquatic ecosys.
- This fig. – small river – aquatic ecosys. – along – mountains, forests, grasslands, farmlands – example – terrestrial ecosys.
- Farmland – **human-made ecosys.**
- All ecosys. – interact – each other – given pt.
- Above act. – importance – components, interactions – ecosys.
- Ex. – sunlight, CO_2, water – essential – produce food – plants – soil – nutrients – plant growth – air – oxygen – respiration – plants, animals – water – essential – all living org.
- This show – living – depend – non-living component – ecosys.
- Biotic – depend – abiotic components – similar – abiotic – influenced by biotic components
- Ex. – plants – release oxygen – photosynthesis – roots – hold soil – prevent erosion – retain moisture – cool atm.
- Identify, study – ecosys. – surroundings – obs. diff. types – interactions – biotic, abiotic components
- Study – biotic interactions – notice – org. – depend – each other – food
- Activity –
  - Obs. – forest ecosys.



- Study pic. – spot org. – listed –
  - Deer, hare, vulture, Bengal fox, bird (*shikra*), squirrel, mouse, mushroom, tree
- Use – internet, school library – find out – org. – eat what?
- Record obs. – identify – each org. – feed – plant products, animals, both

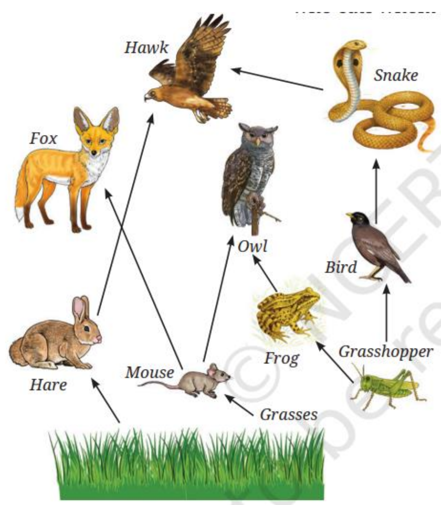
- Plants – make own food – photosynthesis – **producers** OR **autotrophs** (*auto* = self – *troph* = food)
- Org. – cannot produce food – depend – other org. – food – **consumers** OR **heterotrophs** (*hetero* = other – *troph* = food)
- Org. – eat plants – **herbivores** – deer, hare – eat animals – **carnivores** – leopard – eat both – **omnivores** – crows, foxes, mice

Who Eats Whom?



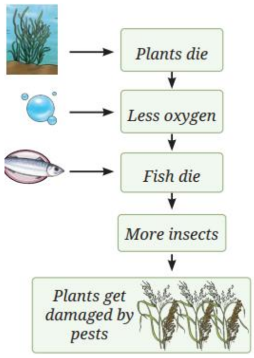
- Above act. – learn – feeding rel. – org. – given ecosys.
- Activity –
  - Take ex. – grassland ecosys.
  - Consider – following org. – spot able – grassland ecosys. –
    - Grass, frog, hare, grasshopper, snake, eagle
  - Fig. – rel. – who eats whom – some org.
  - **Draw** – feeding rel. – remaining org. – add arrows
- Fig. – grass – eaten by hare – eaten by leopard
- This – representation – food chain – grassland ecosys.
- 1 ex. – other chain –
  - Grass → grasshopper → frog → snake → eagle
- Interactions – biotic components – based – feeding rel. – represented – linear chain
- **Food chain** – simple sequence – who eats whom – ecosys.
- Activity –
  - Fig. – crop field – millets, mouse, eagle
  - **Count** – no. – each type – org.
  - Make table – set no. – each org.
  - Arrange no. – ascending order – highest no. – base – lowest no. – top
  - Place – mouse, millet, eagle – appropriate places
  - Looks like – pyramid – complete

- Each org. – food chain – specific pos. – **trophic level** (*troph* = food)
  - Producers – green plants – 1st level
  - Herbivores – hares, deer – 2nd level
  - Small carnivores – frogs – 3rd level
  - Large carnivores – ligers, vultures – 4th level
- Activity –
  - Look – fig.
  - Put arrows – missing rel. – who eats whom
  - Many org. – connected – 1 org. – feeding rel. – ecosys.
- Obs. rel. – diff. food chains – ecosys. – interlinked
- Each org. – eaten – 2 or more org.
- Ecosys. – food chain – interlinked – each other – form network – **food web**



- Living org. – grow – perform many func. – develop – die
- Life cycle – org. – produce waste – dead matter, food waste

What Happens to Waste in Nature?



- May see – umbrella-like structures – mushrooms – grow on – dead plants, trees – rainy season
- These – fungi – grow – dead matter
- Microorganisms – fungi, bacteria – break down – complex substance – dead plants, animals – simple ones
- This process – return nutrients – soil
- Also find – tiny insects – beetles, flies – animal droppings – elephant dung – help – break down – recycle nutrients – back to env.
- This process – **decomposition** – org. – carry process – **decomposers** OR **saprotrophs** (*sapro* = rotten – *troph* = food)
- Plants – grow – soil – many nutrients – origin – decomposition
- Decomposers – imp. role – recycle nutrients
- Nature – nothing wasted – everything reused
- **Interesting fact** –
    - India – country – diverse habitats, seasons
    - **Migratory birds** – fly – 1000s miles – reach diff. habitats – India
    - Migrate from – diff. parts – world – avoid – harsh climate – search – food
    - Birds – enhance aesthetics – habitats – BUT also – significant role – keep balance – ecosys. – **pollinators** – seed dispersers – path – migrate
    - This way – link – 2 habitats
    - Birds – **predators** – insect pests – help farmers – healthy crop growth
    - Migratory birds – Demoiselle Crane – visit – water body – Khichan village – Jodhpur dist. – winter months
    - **Collect** – postal stamps, covers – migratory birds – released – Indian Postal Dept. – collect info. – origin place, reason – migration – diff. loc.
    - **Showcase** – postal stamps – science lab., library – popularize migratory birds

How Does One Change Lead to Another?

- Look – fig. – shows – small change – lead – many others
- Ex. – many plants – pond – start dying – pollution
- Fewer plants – less oxygen – drop – fish population
- Reduction – fish population – cascading effect – less no. – consumers – pond
- Result – insect population – increase – spread – nearby farms – farmers – compelled – use pesticides – grow crops – affect env. further
- Further consequences – emerge – **environmental issues**
- Activity –
    - 1980s – India – significant exporter – frog legs – Indian bullfrog (*Hoplobatrachus tigerinus*)
    - Large-scale export – result – decline – frog populations
    - Frogs – eat insects – populations – increase – forced farmers – more synthetic pesticides – harm env., soil, water quality – affect overall health – env., human
    - Indian govt. – banned export – frog legs – prevent further **ecological damage**

- Ecosystem – in balance – interactions – org., env. – keep resources stable
- This balance – dynamic – can disrupt – natural, human-made changes

How Do Interactions Maintain Balance in Ecosystems?

- Feeding rel. aside – org. – also compete – common resources – food, water, phy. space, sunlight
- This competition – control population size – keep ecosystem balance
- Other types – rel. – ex. given – fig.
- **Mutualism** – both org. benefits – ex. – honeybees, flowers
- **Commensalism** – 1 org. – benefit – BUT – other – not affected – ex. – orchids on trees
- **Parasitism** – 1 org. – benefit – BUT – other – harmed – ex. – ticks – body – dogs
- Interactions – part – complex web – life – ecosys.

What Are the Benefits of an Ecosystem?

- Learnt – biotic, abiotic components – ecosys. – depend – each other – support – various life processes
- Humans – benefit – ecosys. – ex. – fresh air, fertile soil, food, fibres, timber, medicines
- Aquatic ecosys. – water, food – also offer – aesthetic, recreational value
- Support well-being – show – close connection – nature, humans
- Overuse, misuse – natural resources – disturb balance – nature
- Real life ex. – threatened ecosys. – Sundarbans
  - Sundarbans – largest mangrove forests – world
  - Loc. – Ganges, Brahmaputra rivers – meet b/w – India, Bangladesh
  - Sundarbans' forests, rivers – home – various flora, fauna – many – endangered
  - Sundarbans – protect us – slow down – string winds, waves – storms, floods
  - Trees – also absorb – CO_2 – air – release O_2
  - Its imp. – United Nations Educational, Scientific and Cultural Organisation (UNESCO) – declare Sundarbans – World Heritage Site – 1987
  - Sundarbans – under threat – Mangrove trees – cut – fuelwood, farming
  - Illegal hinting, overuse – forest resources – threat – wildlife
  - Pollution – industrial waste, untreated sewage – damage – water, habitat
  - Human act. – disrupt – natural way – ecosys.
- Similarly – other ecosys. – across India – under threat
- Problems – deforestation, overuse – natural species – spread – invasive species – pollution – damage – forest, river, scrubland, wetland, grassland, coastal areas
- **Interesting fact** –
  - Protected areas – parts – land, water – set aside – conserve wildlife, habitats
  - India – many protected areas – national parks, wildlife sanctuaries, biosphere reserves, community conserved areas
  - These places – protect – entire habitats – endangered birds, animals, rare plants
  - Famous ex. –
    - Jim Corbett National Park – Uttarakhand
    - Manas National Park – Assam
    - Nilgiri Biosphere Reserve – Western Ghats
    - Chilika Lake – Odisha
    - Eaglenest Wildlife Sanctuary – Arunachal Pradesh
    - Hemis National Park – Leh

        - Keibul Lamjao National Park – Manipur
        - Pirotan Island Marine National Park – Gujarat
    - Protected areas – big role – save nature – future gen.

Human-made Ecosystem

- Humans – create – artificial ecosys. – fish ponds, farms, parks – needs
- Well designed – help – reduce pollution – support biodiversity – provide recreational space – people
- Unlike natural ecosys. – need – human care, mgmt.

How do healthy ecosystems serve our farms?

- Farming – major livelihood – India – can go unsustainable – not managed – env. friendly tech.
- Humans – practice farming – 1000s yrs. – grow food
- Population increase – dependence – agriculture – increase
- 1950-1965 – India – food crisis – low crop production
- Mid-20th cent. – use – tractors, machines, synthetic fertilisers, pesticides – help – increase production – this period – **green revolution**
- These methods – now – unsustainable – reason – overuse – synthetic chem. – excessive extraction – groundwater – no crop rotation – commercial gain
- Scientists – believe – overuse – pesticides – grow – same crop – lead – soil degradation
- Understand ecosys. – help – adopt – better sustainable practices
- Activity –
    - Visit – nearby farm – parents – teacher – interact – farmers – comm. – find farming practice
    - **Prepare** – list – ques. – farmers – find – pesticides, farm i/p – used – whether – reuse, recycle – improve crops
        - How – farming practices – change – over time? – why?
        - What effects – notice – synthetic pesticides, fertilisers – used?
        - Obs. – any changes – soil health – using synthetic fertilisers, pesticides?
    - **Interact** – farmers – based – these ques. – make report
- Synthetic fertilisers, pesticides – vital role – improve – crop production – help countries – food secure
- BUT – long-term use – affect – env. – soil health
- Overuse – reduce fertility – decrease microbes – lower humus – help bind soil
- w/o humus – soil – prone – erosion – also reduce – population – natural predators – increase population – pests
- Heavy irrigation – repeated ploughing – also disturb – soil org. – earthworms, snails – imp. – maintain eco. balance
- Some pests – develop resistance – pesticides – difficult – control
- Grow – same crop – on repeat – **monoculture** – reduce biodiversity – affect pollinators – crucial – food production
- Farming – more sustainable – farmers – explore – organic, natural methods
- Aim – reduce use – synthetic fertilisers – minimal interference – natural ecosys.
- **Interesting fact** –
    - Ancient text – *Vrikshayurveda* – emphasis – soil health, nourishment
    - Text – advocate – cont. nourishment – soil – organic manure – *Kunapa Jala* (liquid fertiliser – made – animal, plant waste – process – fermentation – breaks – complex substance – simple ones